

BITCOIN

Market Sentiment vs Trader Performance

An Analytical Study on Hyperliquid Trading Data & the Fear & Greed Index

PrimeTrade.ai — Data Science Hiring Assignment

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Sentiment-Driven Insights for Adaptive, Data-Backed Trading Strategy

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1 Executive Summary

This project evaluates whether Bitcoin market sentiment, measured using the Fear & Greed Index, influences trader performance on the Hyperliquid exchange. Historical trade data was merged with daily sentiment labels to study profitability, trading behaviour, trader performance, and asset-level performance.

KEY RESULT

Statistical testing confirmed that market sentiment is associated with significant differences in average trader profitability.

2 Dataset Overview

Historical Trader Data

Account, Coin, Execution Price, Size Tokens, Size USD
Side, Direction, Closed PnL, Fee, Timestamp and identifiers

Fear & Greed Index

Daily sentiment classification: Extreme Fear, Fear, Neutral, Greed, Extreme Greed
Accompanying daily sentiment score

After preprocessing, both datasets were merged on the trading date to enrich every trade with the prevailing market sentiment.

3 Methodology

Data cleaning included timestamp conversion, datatype correction, removal of missing sentiment values, and feature engineering.

Engineered Features

- IsWin — binary flag for profitable trades
- IsLoss — binary flag for loss-making trades
- Absolute PnL — magnitude of profit or loss
- Trade Value — total value of the executed trade
- PnL Percentage — return relative to trade value

Analytical Scope

Exploratory analysis included sentiment distribution, trader analysis, coin analysis, profitability trends, correlation analysis, and statistical hypothesis testing.

4 Exploratory Data Analysis

The following analyses were performed to understand trading behaviour and its relationship with market sentiment:

✓ Market Sentiment Distribution	✓ Average & Total Closed PnL by Sentiment
✓ Win Rate Comparison	✓ BUY vs SELL Performance
✓ Direction Analysis	✓ Top and Bottom Performing Traders
✓ Coin-wise Profitability	✓ Correlation Heatmap
✓ Daily Profit Trend	✓ Fee vs Profit and Trade Size vs Profit

5 Statistical Analysis

One-Way ANOVA

Metric	Result
Test	One-Way ANOVA
F-statistic	9.062
p-value	< 0.001
Conclusion	Null hypothesis rejected — sentiment groups differ significantly

Tukey HSD — Pairwise Comparisons

Significant pairwise differences were identified between the following sentiment category pairs:

Group A	Group B	Result
Extreme Fear	Extreme Greed	Significant
Extreme Greed	Greed	Significant
Extreme Greed	Neutral	Significant
Fear	Neutral	Significant

6 Key Findings

- Market sentiment significantly influences trader profitability.
- A small number of traders contribute disproportionately to cumulative profits.

- Cryptocurrency selection impacts profitability.
- Trade size is positively correlated with transaction fees.
- Closed PnL has weak linear correlation with individual numeric variables, indicating profitability depends on multiple interacting factors.
- Statistical testing validated that observed differences are unlikely to occur by chance.

7 Business Recommendations

RECOMMENDATION 1

Use the Fear & Greed Index as an input to trading strategies.

RECOMMENDATION 2

Reduce exposure during Fear and Extreme Fear markets.

RECOMMENDATION 3

Benchmark consistently profitable traders.

RECOMMENDATION 4

Prioritize assets with stable profitability across sentiments.

RECOMMENDATION 5

Combine market sentiment with trader-level analytics for stronger decision-making.

8 Conclusion

The analysis demonstrates that market sentiment provides valuable information about trading performance but should be combined with trader behaviour, risk management, and asset selection. Statistical validation strengthens confidence in these findings and supports the development of adaptive, sentiment-aware trading strategies.

— End of Report —